T1D PRAJANA YANDRA

Paediatric Enhancement mHealth App (T1DPE mApp)

Business Requirements Document

Playbook · Open Questions · Analytics Plan · Application Specification

Field	Detail
Document status	Draft v1.0 — for stakeholder review
Date	11 August 2026
Prepared for	teammistaketechnologies@gmail.com
Source material	Google Drive export: drive-download-20260811T152110Z-1-001.zip (17 files — 16 content docs + 1 PhD compliance proposal PDF)
Engineering status	Backend (Next.js API + Admin console) implemented against a Prisma/PostgreSQL schema. Flutter mobile app is currently an unmodified framework scaffold — no screens built yet.
Research anchor	PhD (Nursing) study, The Tamil Nadu Dr. M.G.R. Medical University — Mrs. Mahalakshmi N.B., KMCH College of Nursing, Coimbatore. Guide: Dr. Vijayalakshmi.

0. How to read this document

This BRD turns the Google Drive export and the existing codebase into a single reference for what the T1D PRAJANA YANDRA app is, what it must do, and what it takes to ship it on the App Store and Google Play. It is organised as:

1. Research context — the PhD study this app exists to serve, and why that shapes every other section.
2. Source content inventory — every file in the drive export, decoded and mapped to app modules.
3. Current build status — what is already implemented in this repository vs. what is still a scaffold.
4. Application playbook — personas, every existing admin page, and the proposed mobile app page list with features.
5. Feature flags — the criteria and the recommended flag list.
6. Questionnaires & analytics plan — how the research instruments map to data captured in-app and to the statistical plan.
7. Store readiness — the signed claims and compliance artifacts required for Google Play and the Apple App Store, given this is a children's health-data app tied to a clinical study.
8. Open questions — decisions we need from you before build-out continues.
9. Next steps / roadmap.

1. Research context — why this app exists

The content in the drive export is not generic wellness copy. It is the intervention material for a PhD (Nursing) study, and the mobile app is the study's delivery vehicle for a 4–6 week follow-up phase after in-person teaching. Every product decision below (content structure, bilingual requirement, assessment forms, experimental/control cohorts) traces back to this proposal.

1.1 Study identity

Item	Value
Study title	Effectiveness of T1D PRAJANA YANDRA Paediatric Enhancement mHealth App (T1DPE mApp) on Physical Well-being of Children and Parental Satisfaction among children with type 1 diabetes mellitus in selected Diabetic clinics at Coimbatore
Institution	The Tamil Nadu Dr. M.G.R. Medical University, Chennai
Scholar / Candidate	Mrs. Mahalakshmi N.B., Professor, KMCH College of Nursing, Kalappatti, Coimbatore
Guide	Dr. Vijayalakshmi, Principal, Vignesh Nursing College, Tiruvannamalai
Clinical setting	Maadhuram Diabetic Center, Coimbatore
Design	Mixed methods — quantitative quasi-experimental (pretest–posttest, experimental vs. control) + qualitative descriptive (parent interviews)

Item	Value
Sample	140 children with T1D (70 experimental / 70 control), ages 6–15, consecutive sampling
Inclusion criteria	Age 6–15, either gender, diagnosed 6–12 months prior, parent able to give assent, parent owns an Android phone
Exclusion criteria	Children with DKA or hyperosmolar hyperglycaemic non-ketotic syndrome complications

1.2 Objectives and hypotheses

10. Assess physical well-being of children with T1D and their parents' satisfaction.
11. Evaluate the effectiveness of the T1DPE mApp on physical well-being and parental satisfaction, experimental vs. control.
12. Correlate physical well-being with parental satisfaction in both groups.
13. Test association between demographic variables and pretest physical well-being / parental satisfaction.
14. Explore the lived experience of parents in both groups (qualitative).

> *H1: significant difference in physical well-being / parental satisfaction, experimental vs. control. H2: significant correlation between physical well-being and parental satisfaction. H3: significant association between demographics and pretest scores. Significance level: α = 0.05.*

1.3 Intervention structure — the 8 sessions

Delivered face-to-face for weeks 1–4 (twice weekly, 30–40 minutes), then reinforced via the app for weeks 4, 5 and 6. This session table is the backbone of the mobile app's content module list in Section 4.3.

Wk	Session	Topic	In-person method
1	1	Techniques of administering insulin injection or pump	Demonstration
1	2	Blood glucose monitoring (SMBG)	Demonstration
2	3	Hypoglycaemia — identification, management, prevention	Video
2	4	Planning and monitoring nutrition	Pamphlet
3	5	Muscle and bone strengthening exercises	Video
3	6	Management in school	Pamphlet
4	7	Travel tips	Pamphlet
4	8	Diabag (diabetes supplies kit)	Video

> *"Physical well-being" is operationally defined for this study as glycemic control + HbA1c + PedsQL-measured quality of life. "Parental satisfaction" is measured with the WE-CARE questionnaire. Both are captured pre- and post-intervention — see Section 6.*

2. Source content inventory (drive export)

The zip contains 16 content documents, each supplied as an English/Tamil pair (identical structure, translated body copy), plus one compliance PDF. This confirms the app must ship fully bilingual (English + Tamil), not just localized UI chrome — the clinical education body copy itself needs a translation pipeline.

Source file (paired EN/TA)	Session	Maps to module	Notes
InsulinTypesSto*.docx	1	Insulin — Types & Storage	3 articles: insulin types/colour codes, injection technique & sites, insulin regimen (MDI, IC/ISF formulas) + quiz
insulin pump*.docx	1	Insulin — Pump	Pros/cons, basal-rate rules (dawn/dusk phenomena), FAQ
SMBG*.docx	2	Glucose Monitoring	Glucometer selection & technique, CGM/FGM (Libre) FAQ, ADA target ranges, true/false quiz
hypoglycemia*.docx	3	Hypoglycaemia	Causes/symptoms, severity levels, treatment (rule of 15), glucagon dosing table, prevention, sequencing quiz
Nutrition*.docx	4	Nutrition	Carb counting (IC/IS ratio worked examples), school tiffin guidance, 8 recipes, milk (A1/A2), fruit guidance, celiac/gluten-free diet, FAQ
EXERCISEQUIZ*.docx	5	Exercise	Short Q&A: sports/BSL management, high-altitude trekking
Type1 DM &travel care*.docx	7	School & Travel	Also covers school + travel checklist, sample doctor's travel certificate, packing list
annual checkup*.docx	n/a (supporting)	Annual Check-up / Tests	Ketone testing, HbA1c, annual screening schedule (thyroid, celiac, lipid, UACR, retina)

Source file (paired EN/TA)	Session	Maps to module	Notes
PhD Questionnaire (corrected)*.docx	n/a	Assessment instrument	Tamil-language structured tool: demographics (Sections A/B), physical-parameter pre/post table (Section C) — feeds the research module, not education content
N B Mahalakshmi proposal for compliance.pdf	n/a	Study protocol	27-page PhD proposal — objectives, instruments (PedsQL, WE-CARE), sample size, statistics plan (see Section 6)

> *"Management in school" (Session 6) and "Diabag" (Session 8) are covered inside the travel-care and nutrition documents rather than as standalone files — flag to the content owner in case dedicated files exist but weren't included in this export (see Open Questions, Section 8).*

3. Current build status

3.1 What exists today

Per the git history ("Bootstrap Next.js API and Flutter app scaffolds" → "feat(api): session-based routing guard for admin pages" → "backend done"), the repository has two apps:

- api/ — Next.js 15 app. A full data model (Prisma/PostgreSQL, 1,200-line schema) plus a working REST API and an internal Admin console (staff-only, session-guarded). This is functionally complete for staff/researcher workflows.
- app/ — Flutter mobile app. Still the default "create Flutter app" counter-button scaffold (lib/main.dart, 122 lines). No patient-facing or parent-facing screens exist yet.

> *Read literally, "init setup is done" refers to the backend/admin half only. The patient/parent mobile app — the actual delivery vehicle for the 8 sessions and the follow-up weeks 4–6 — is 0% built. This BRD's Section 4.3 mobile page list is therefore the spec for that build, not a description of existing screens.*

3.2 Data model coverage (already built)

The Prisma schema already anticipates nearly everything this BRD calls for:

Domain	Existing models
Identity & roles	User (PATIENT / ADMIN / SUPER_ADMIN / RESEARCHER / CLINICAL_REVIEWER), Profile, AdminUser, Session/Account (Better Auth)
Glucose & HbA1c	GlucoseReading (context-tagged: fasting/pre-meal/post-meal/bedtime/overnight/exercise), HbA1cRecord
Insulin	InsulinLog (type, dose, injection site, meal association) — logging only, app never recommends a dose

Domain	Existing models
Nutrition	Meal, MealItem (carb/calorie/macro fields ready for carb-counting workflow)
Exercise	Exercise catalogue + ExerciseLog + ExerciseContent (video/instructions/precautions)
Medication	Medication + MedicationLog + Reminder
Extensible metrics	HealthMetricDefinition + HealthMetric — new measures added as data, not schema changes
Clinical thresholds	ClinicalThreshold (global/study/participant scope) — no hard-coded clinical cut-offs anywhere in code
Education content	EducationContent (category, rich body, media, status/versioning) — this is where the drive docs land
Research	ResearchStudy, StudyParticipant (arm/group, consent, enrollment), StudyAccess, DataExport, ClinicalThreshold(scope=STUDY)
Notifications	NotificationCampaign, Notification, DeviceToken, Reminder — push infrastructure for session/questionnaire reminders
Governance	AuditLog (append-only), SystemSetting

> *Gap: there is no PedsQL / WE-CARE questionnaire model yet. HealthMetric's generic key/value shape can hold the scored domains (see Section 6.4), but the item-level responses (23 PedsQL items, 37 WE-CARE items) need either a new QuestionnaireResponse model or a JSON field on HealthMetric. Recommend a dedicated model — see Open Questions.*

4. Application playbook

4.1 Personas

Persona	Role in schema	What they need from the app
Child (6–15 yrs)	User(PATIENT), linked Profile	Simple, mostly-parent-operated logging; age-appropriate education content, quizzes
Parent / caregiver	Operates the child's account (Android phone requirement in inclusion criteria)	Session content, reminders, logging glucose/insulin/meals/exercise, WE-CARE questionnaire, Diabag checklist, travel certificate
Clinician / diabetes educator	User(CLINICAL_REVIEWER / ADMIN)	Participant timeline, thresholds, exports

Persona	Role in schema	What they need from the app
Researcher (Mahalakshmi & team)	User(RESEARCHER)	Study setup, arm assignment, pre/post questionnaire data, statistical exports
Content admin	User(ADMIN)	Upload/translate/publish the 16 documents as EducationContent + ExerciseContent, manage notification campaigns

4.2 Admin web console — pages that exist today (api/app/admin)

31 pages, session-guarded, already implemented:

Page	Purpose
Auth: login / forgot-password / reset-password / no-access	Staff authentication (Better Auth), role-gated access denial screen
Dashboard	Landing overview for staff
Participants (list + [id] detail)	Roster + individual participant profile
Health → Glucose / Insulin / HbA1c / Meals / Medications / Exercise / Metrics	Per-domain data browsers across all participants
Content → Education / Exercises	CMS for EducationContent and ExerciseContent — this is where the 16 drive docs get entered
Research → Studies (list + [id]) / Export	Study configuration, arm/cohort management, data export requests (CSV/XLSX/PDF)
Thresholds	Clinician-configured clinical cut-offs (global/study/participant scope)
Notifications	Campaign creation/scheduling/sending, targeted by role/study/user list
Reports	Aggregate reporting views
Administrators	Staff user management
Audit logs	Append-only action trail
Settings / Account	System settings, staff account management

4.3 Mobile app — proposed page list (Flutter, patient/parent-facing) — TO BE BUILT

Mapped directly to the 8 sessions plus the assessment instruments. This is the spec for Phase 2 build-out, not existing screens.

Screen	Features	Source module
Onboarding & consent	Language picker (EN/TA), study arm-blind enrollment, guardian assent capture, Profile creation	Study protocol (informed consent)
Home / Today	Session progress (1 of 8), streak, upcoming reminders, quick-log shortcuts	—
Insulin — Types & Injection	Article reader (color-coded insulin chart), injection-site diagram, quiz, log an insulin dose	InsulinTypesSto
Insulin Pump	Pros/cons article, basal-rate rule reference card, FAQ accordion	insulin pump
Glucose Monitoring (SMBG)	Glucometer how-to, CGM/Libre FAQ, target-range reference (from ClinicalThreshold), log a glucose reading, trend chart	SMBG
Hypoglycaemia	Severity flowchart, "rule of 15" calculator, glucagon dosing card, prevention tips, emergency contact quick-dial	hypoglycemia
Nutrition	Carb-counting tutorial + IC/ISF calculator, school tiffin planner, recipe library (8 recipes), celiac/gluten-free guide, log a meal	Nutrition
Exercise	Muscle/bone-strengthening video library, pre-exercise BSL checklist, log a workout	EXERCISEQUIZ + ExerciseContent
School Management	Teacher/school-letter template, hypo kit checklist for school bag, tiffin timing table	Type1 DM & travel care
Travel & Diabag	Pre-travel checklist, printable travel certificate, packing list generator, cold-chain tips	Type1 DM & travel care
Annual Check-up Tracker	Ketone-testing how-to, due-date tracker for HbA1c/thyroid/celiac/lipid/UACR/retina screening	annual checkup
My Log / Records	Glucose, insulin, meal, exercise, HbA1c history — read view of the same data the admin console shows staff	Existing API
Assessments	PedsQL (23 items) and WE-CARE (37 items) forms at pretest and posttest; demographic/clinical intake form	PhD Questionnaire + proposal PDF

Screen	Features	Source module
Reminders / Notifications	Session unlocks, questionnaire due dates, medication/glucose check reminders	Existing Reminder/Notification models
Profile / Settings	Language toggle, child profile, notification preferences, data export/delete request, privacy policy, consent withdrawal	Compliance requirement

> *Each content screen should render a Quiz component (all 8 source docs end in a quiz/FAQ) that logs completion — this is the natural place to capture "session viewed" events for the researcher's adherence tracking, and it doubles as the child-facing engagement mechanic.*

5. Feature flags

5.1 Criteria for adding a flag

15. Gates anything shown differently to the experimental vs. control study arm (control gets "routine care" — i.e., app content may need to be hidden or limited for control-arm accounts if the protocol requires it).
16. Gates content still pending translation — publish English, hold Tamil (or vice-versa) per document without blocking the other.
17. Gates anything requiring app-store review lead time (e.g., HealthKit/Health Connect sync) so it can ship dark and be turned on post-approval.
18. Gates a clinical safety-sensitive feature (glucagon dosing calculator, IC/ISF calculator) behind a clinician sign-off toggle per study site.
19. Gates a paid/regional feature (e.g., CGM integrations like Libre/Dexcom) that not all participants' clinics support.
20. Supports kill-switch rollback of any notification campaign type without an app-store re-release.

5.2 Recommended flag list

Flag	Default	Purpose
study_arm_control_routine_care_only	off	If the protocol requires the control arm to see fewer/no app modules, this suppresses Sessions 1–8 content for control-group accounts
content_lang_ta_enabled	off until QA'd	Per-document Tamil content publish switch, independent of English
cgm_integration_enabled	off	Libre/Dexcom/Glimp-style device sync — heavier store review and device-compatibility burden

Flag	Default	Purpose
glucagon_dose_calculator	off	Age/weight-based glucagon dosing helper — clinical-safety review gate per site
ic_isf_calculator	off	Insulin:carb and correction-factor calculator — same clinical-safety gate
pedsql_posttest_reminder	on	Automated posttest questionnaire nudge at week 6
push_notifications_enabled	on	Master switch for the NotificationCampaign pipeline
data_export_self_service	off	Lets a parent request/download their own child's data (DPDP/GDPR-style right of access) ahead of full deletion-flow build

> *Implementation note: SystemSetting already exists as a generic key/value store scoped by category — flags can live there today without a schema change, or promote to a dedicated FeatureFlag model if per-study or per-cohort targeting is needed.*

6. Questionnaires & analytics plan

6.1 Assessment instruments (from the PhD proposal)

Section	Instrument	Detail
A	Demographics — child	Age, sex, family type (nuclear/joint/extended), birth order, residence (rural/urban), family history of diabetes, information source
A (cont.)	Demographics — parents	Parent education level, occupation, family income band
B	Clinical variables	Weight, age at diagnosis, illness duration, consanguinity/family history, medication type, diet pattern, fast-food frequency, FBS/PP glucose, HbA1c
C	Structured observational tool	Anthropometric data (height, weight, BMI) pre-test and post-test
D	PedsQL (Pediatric Quality of Life Inventory)	23 items, 5-point Likert (0–4), 4 domains: Physical, Emotional, Social, School Functioning. Score bands: Poor 0–30 (<33%), Moderate 31–61 (34–66%), Good 62–92 (>67%)

Section	Instrument	Detail
E	WE-CARE (Well-being and Satisfaction of Caregivers of Children with Diabetes)	37 items, 4 domains: Psychosocial Well-being (13), Ease of Insulin Use (9), Treatment Satisfaction (9), Acceptance of Insulin Administration (6). Score bands: Low 0–49 (<33%), Average 50–98 (34–66%), High 99–148 (>67%)
Qualitative	Semi-structured interview (parents only)	Manual coding, descriptive thematic analysis — collected outside the app, not an in-app form

6.2 Data capture → statistical plan mapping

Study objective	Statistical method	Data source in-app
Assess physical well-being & parental satisfaction	Frequency & percentage distribution	PedsQL + WE-CARE scored responses, both arms, pretest
Evaluate T1DPE mApp effectiveness (experimental vs. control)	Mean & SD, ANOVA	PedsQL/WE-CARE pre vs. post, GlucoseReading + HbA1cRecord trends, both arms
Correlate well-being with parental satisfaction	Pearson correlation	Paired PedsQL and WE-CARE scores per participant
Associate demographics with pretest scores	Chi-square	Section A/B demographic fields vs. PedsQL/WE-CARE score bands
Explore parent lived experience	Descriptive thematic analysis	Interview transcripts — collected outside the app; store only a reference/consent flag in-app

> *None of these tests need to run inside the app. The app's job is clean capture (structured forms, not free text, wherever the instrument is a scale) plus a DataExport (already modeled) that a statistician runs through R/SPSS. The admin Research → Export page should offer a "pretest/posttest paired dataset" export type specifically for this study.*

6.3 Admin analytics/reporting — what should be on screen

- Study-level dashboard: enrollment vs. target (140), arm balance (70/70), consecutive-sampling audit trail.
- Adherence: % of assigned sessions viewed per participant, weeks 4–6 app engagement (login frequency, module completion) — this is the "effectiveness" proxy the app itself can measure in real time, distinct from the clinical PedsQL/WE-CARE outcome.
- Glycemic trend charts per participant and per arm (GlucoseReading, HbA1cRecord already time-indexed for this).
- Questionnaire completion tracker: pretest done / posttest due / posttest done, per participant.

- Export queue: CSV/XLSX for SPSS import, mapped to participantCode / studyParticipantCode pseudonyms only — never raw name/contact fields (already the schema's design).

6.4 Data model gap to close before Phase 2

Add a QuestionnaireResponse (or similarly named) model: participantId, studyId, instrumentKey (PEDSQL | WE_CARE), phase (PRE | POST), itemResponses (Json — item code → Likert value), domainScores (Json), totalScore, scoreBand, completedAt. This keeps raw item-level data auditable for the statistician while HealthMetric continues to hold the physical measurements (height/weight/BMI/glucose/HbA1c) already modeled.

7. Store readiness — required signed claims & compliance

This app collects sensitive health data about children, and is tied to a formal research study with informed consent. That combination raises the compliance bar on both stores well above a typical consumer health app. Everything below should be treated as a submission blocker, not a nice-to-have.

7.1 Google Play — required declarations

Requirement	What must be filed / signed
Data safety form	Must disclose collection of health info, physical activity, personal identifiers, and (if children are direct users) that data may relate to a child. Declare purpose ("App functionality", "Account management") — not advertising.
Target audience & content declaration	Must declare the app is used by / on behalf of children (6–15). This triggers Google Play Families Policy review even though the parent operates the account — the child is a data subject.
Families Policy compliance (if in scope)	No behavioral ads, no third-party analytics/ad SDKs without a child-directed exemption, restricted permissions, privacy policy must have a dedicated child-data section.
Health Connect / sensitive health data policy	If integrating CGM/glucometer device data via Health Connect, self-certify compliance with Google's Sensitive Health App policy before that feature flag is turned on.
Permissions declaration	Justify any BODY_SENSORS, notifications, storage, or Bluetooth (device pairing) permission in the Play Console permissions declaration form.
App signing	Enroll in Play App Signing; retain the upload key separately from Google-managed signing key; document key custody (who holds the upload keystore) for continuity if the original developer leaves.
Privacy Policy URL	Publicly hosted, must cover: what's collected, retention period, research use, right to withdraw consent/delete data, contact for data requests. Must be linked in both the Play listing and in-app.

Requirement	What must be filed / signed
Content rating (IARC)	Complete the IARC questionnaire — expect a rating suitable for children given no ads/violence, but must reflect health-content disclaimers.

7.2 Apple App Store — required declarations

Requirement	What must be filed / signed
App Privacy "Nutrition Label"	Declare data types: Health & Fitness (blood glucose, insulin doses, exercise, HbA1c), Contact Info (parent), and whether data is "linked to identity" — it is, via Profile/participantCode which is pseudonymous but re-identifiable by staff.
Kids Category rules (if listed under Kids)	If listed in the Kids category: no third-party analytics/ads, no external links without a parental gate, COPPA-aligned data minimization. Given the parent operates the account and consents, evaluate listing as a regular Medical/Health app with age rating 4+/9+ instead — avoids the stricter Kids-category ad/analytics ban conflicting with research analytics needs (confirm with legal — see Open Questions).
HealthKit entitlement (if used)	Requires the HealthKit capability + usage strings (NSHealthShareUsageDescription / NSHealthUpdateUsageDescription) and Apple's HealthKit review — cannot be used for advertising or data-broker purposes; must state research use plainly.
Export compliance	Declare encryption usage (ITSAppUsesNonExemptEncryption in Info.plist) — standard HTTPS/TLS only qualifies for the exempt declaration; file it either way at each submission.
Sign in with Apple	Required only if a third-party social login (Google, Facebook, etc.) is offered — plan the auth provider list now to avoid a late addition forcing this.
Distribution certificate & provisioning	Apple Distribution certificate + App Store provisioning profile tied to the org's Team ID; document who holds the private key (Keychain/CI secret) for release continuity.
Privacy Policy URL	Same content requirement as Google Play; must be reachable from the App Store listing and Settings screen in-app.
Age rating questionnaire	Complete App Store Connect's age rating survey reflecting medical/treatment information content.

7.3 Cross-cutting compliance (both stores, plus research ethics)

Area	Requirement
Institutional/ethics approval	IRB/Institutional Ethics Committee approval number for the study should be referenced in-app on the consent screen (schema already has ResearchStudy.irbNumber and consentVersion — surface it in the UI, don't leave it admin-only).
Informed consent + assent flow	Parent consent AND age-appropriate child assent, versioned (consentVersion field exists) — must be re-shown if consent text changes.
India DPDP Act 2023	As the population is India-based: lawful basis for processing a child's personal/health data, parental consent verification, data localization/retention limits, and a grievance officer contact — needed regardless of app-store requirements.
Data retention & deletion	Document and implement a retention schedule; support consent withdrawal (StudyParticipant.withdrawnAt exists) with a defined data-handling outcome (retain de-identified vs. delete).
No unlicensed medical advice	Insulin dosing/glucagon calculators (Section 5 flags) need a clear "educational aid, confirm with your clinical team" disclaimer — both stores scrutinize apps that appear to prescribe treatment.
Security baseline	Encryption in transit (TLS) and at rest for health tables, role-based access already modeled via UserRole — confirm production secrets/DB are not shared with any non-production environment.

> *Recommendation: none of the store submission work should start until the Section 6.4 QuestionnaireResponse model and the mobile app (Section 4.3) exist — but the privacy policy, consent copy, and IRB reference should be drafted in parallel now, since legal review lead time is usually the longest pole in the tent.*

8. Open questions

Decisions needed from you (or the research team) before Phase 2 build-out proceeds:

21. Control-arm app access: does the protocol require the control group to see zero app content ("routine care" only), or do they get the app after the study concludes? This drives the study_arm_control_routine_care_only flag default.
22. "Management in school" and "Diabag" (Sessions 6 and 8) don't have dedicated source files in this export — are there separate documents to send, or should we author them by extracting the relevant passages already embedded in the travel-care and nutrition documents?
23. Tamil is confirmed for all 8 education documents — should the mobile app UI chrome (buttons, labels, nav) also ship in Tamil at v1, or English-only chrome with Tamil content initially?
24. Should the IC/ISF calculator and glucagon-dose calculator ship at all in v1, given the store scrutiny on dosing tools, or launch as reference tables only (no calculation) until clinical sign-off is formalized?
25. Who is the data controller of record for DPDP Act / privacy-policy purposes — the university, the clinic (Maadhuram Diabetic Center), or a separate legal entity behind this codebase?

26. Is CGM/Libre device integration (Section 5.2 flag) in scope for this study's 6-week window, or purely a future roadmap item? It materially changes Health Connect/HealthKit review scope.
27. Confirm the target platforms: proposal's inclusion criteria requires an Android phone — is iOS support required at all for study participants, or only for a future public release after the study concludes?
28. For the PedsQL and WE-CARE instruments: do we have a license/permission on file to digitize these copyrighted tools inside the app (both are licensed instruments, not public domain)?
29. Confirm the plan referenced in your message — "we will use admin to upload this data in next phase" — should each of the 16 documents become one EducationContent row per language (32 rows total), or one row with a language toggle field? The current schema has no language column on EducationContent; recommend adding one now to avoid a slug/duplication scheme.

9. Next steps / roadmap

30. Answer Section 8's open questions.
31. Schema addition: add a language field to EducationContent (and ExerciseContent) plus the QuestionnaireResponse model from Section 6.4.
32. Content ops: transcribe the 16 Word documents (32 language variants) into EducationContent via the existing admin Content → Education page; tag each with the correct EducationCategory and sortOrder matching the 8-session sequence.
33. Mobile build: scaffold the Flutter app's real navigation and the screens in Section 4.3, wired to the existing REST API (already implemented — no backend blocker).
34. Assessment forms: build PedsQL and WE-CARE as in-app forms once the QuestionnaireResponse model lands; wire scoring bands per Section 6.1.
35. Feature flags: stand up SystemSetting-backed flags per Section 5.2, default all clinical-calculator flags to off.
36. Compliance: draft privacy policy + consent/assent copy, obtain IRB number for in-app display, complete Play Data Safety form and Apple Privacy Nutrition Label in parallel with mobile build.
37. Pilot: run a small internal cohort through onboarding → session 1–8 → posttest before opening enrollment to the full 140-participant sample.

— End of document —